Tendai Chimuka

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SUMMARY

Electrical Engineering senior with hands-on experience in circuit and PCB design, RF and power electronics, and board bring-up and debugging. Strong foundation in analog/digital circuits and electromechanical design & integration.

EDUCATION

New Jersey Institute of Technology **May 2027**

BS Electrical Engineering *Newark, NJ*

Relevant Coursework: High Speed Electronics (RF), Power Electronics, Electronics Design, Advanced & Feedback Controls

EXPERIENCE

Research Assistant - Electronics Design **Oct 2026 – Present**

NJIT Swissler Innovative Robotics Lab *Newark, NJ*

- Designing circuit subsystems and custom PCBs for modular, self-assembling robot swarms.
- Soldering, assembling, and debugging prototype robot hardware, from board bring-up through mechanical integration.

Research Assistant - Hardware Integration **June 2025 – Present**

NJIT Network & Intelligent Systems Research Lab *Newark, NJ*

- Led a team building energy-efficient 3D stereo vision hardware for autonomous UAVs; paper accepted at MIT URTC.
- Designed and built an autonomous data-collection drone, modeling its structural and payload parts in Creo Parametric, and assembled, soldered, and power-validated its electronics before each flight test.
- Co-authored *EcoFlight: Low-Energy Path Finding Through Obstacles for Autonomous Sensing Drones* (IEEE Xplore).

Research Assistant - Simulations **May 2026 – Present**

NJIT Controls Automation & Robotics Lab *Newark, NJ*

- Designed linkage-driven finger mechanisms and actuator mounts for a robotic hand in Creo Parametric.
- Reviewed grasping and control literature and scoped simulation of finger motion and grasp behavior using MuJoCo.

PROJECTS

NASA RockSat Stella II: CubeSat Flight Computer | *Altium, C++* **Sep 2026 – Present**

- Electrical team manager on a NASA sounding-rocket payload that ejects a cube to measure Earth's magnetic field.
- Designing the cube's flight computer PCB (Teensy 4.1, IMU, ±200 g accelerometer, boom magnetometer over SPI, 2.4 GHz XBee link), taken through preliminary design review and accepted by NASA.
- On the prior payload (Apr 2026), co-designed a 4-layer mixed-signal PCB system for a Langmuir probe and validated its 28V-to-5V buck and 5V-to-3.3V LDO rails, debugging boards at the component level.

5.8 GHz Doppler Radar with PCB Antennas | *openEMS, COMSOL, KiCad* **Sep 2026 – Present**

- Designing a 5.8 GHz Doppler radar with a custom RF front end built from discrete parts (VCO, splitter, low-noise amplifier, and IQ demodulator), kept within FCC limits without a power amplifier.
- Designed and simulated the transmit and receive patch antennas in two different solvers, which agreed within 10 MHz, and chose a low-loss Rogers substrate over FR4 to keep the tuning stable.
- Planned the signal processing so an MCU can measure speed and direction from the IQ outputs with an FFT.

4S Battery Management System with Active Balancing | *KiCad, LTspice, CAN* **Sep 2026 – Present**

- Designing a 4S Li-ion BMS module for electric aircraft around the TI BQ76942, with tiered fault handling, redundant overvoltage protection, a CAN interface, and a 160 A / 60 μs short-circuit trip.
- Designed a bidirectional inductor-based active balancer (100 kHz, ∼1 A/stage), sizing the inductor from first principles.
- Simulated a balancer stage in LTspice, matching hand-calculated peak current, slopes, and ramp times within 4%, and debugged a gate timing fault that caused reverse current.

Tilt-Rotor eVTOL UAV | *Founder & Electromechanical Lead, Senior Capstone* **Sep 2026 – Present**

- Proposed the tilt-rotor eVTOL concept and founded a 6-person capstone team; lead electromechanical design (power, actuation, and airframe) for a 1.1 m span, 2.5 kg vehicle and its 0.7 m 3D-printed subscale testbed.
- Designing a custom power distribution board for 4 tilt servos and the control-surface servos, isolated from the flight controller supply so a servo stall cannot brown out the Pixhawk 6X autopilot.
- Built the mass/CG model and tilt-sweep clearance checks, rebalancing the layout from 64% to 31% of chord and fixing a rear prop-to-tail clash; sized the 6S lift rotors for T/W 2.0 at 285 W hover.

TECHNICAL SKILLS

Simulation & CAD: Altium, KiCad, Multisim, LTspice, openEMS, COMSOL, MATLAB, Simulink, Creo Parametric
Instrumentation: Oscilloscope, Waveform Generator, Multimeter, Power Supply, Spectrum Analyzer, Logic Analyzer
Embedded/Programming: Python, C++, Linux, UART, I^2C, CAN, SPI